



Unit 3 · Outcome 2 · Sustainable Development

Challenges to sustainability

KK04

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



01The core challenge

01 The core challenge Why "do the sustainable thing" is harder than it looks. The six sustainability principles (you met these in KK03) describe what a sustainable society should do — protect ecological integrity, use resources efficiently, be fair to people now and in the future, act cautiously under uncertainty, and make polluters pay. The trouble is that human demand keeps climbing, and four big drivers make every principle harder to honour at once: Population More people means more of every demand — the multiplier behind the other three. Food Fee

02Pressure 1 — Population

02 Pressure 1 — Population The multiplier sitting behind every other pressure. Global population passed 8 billion in 2022 and is still rising, though the rate is slowing. For most of history population grew slowly; in the last two centuries it shot up in a classic exponential ("J") curve as medicine, sanitation and industrial food lowered death rates. Every extra person adds demand for food, water, energy, housing and materials — so population is the multiplier that scales up all the other challenges. Carrying capacity (K) The maximum population size an

03The food–water–energy nexus

03 The food–water–energy nexus Pull one lever and the whole system moves. Drag the slider to see it. The interactive below is the single most important idea in this dot point. The big node is population . The three rings are food, water and energy demand. Raise the population and watch every demand climb together — and notice the linked arrows : food needs water and energy; energy needs water; water delivery needs energy. There is no isolated fix. Interactive — Food · Water · Energy · Population nexus Population 3 billion Food — Water — Energy — Show the

04Pressure 2 — Food

04 Pressure 2 — Food Feeding more people from a fixed, fragile land base. Roughly half the world's habitable land is already farmed , and agriculture uses about 70% of all freshwater withdrawals . Feeding a growing population pushes farming in two damaging directions: extensification (clearing more habitat for cropland and grazing) and intensification (more fertiliser, irrigation, pesticides and machinery per hectare). Both strain the principles. Food pressure What happens Principle challenged Land clearing for crops/grazing Habitat and species lost; carbon released Ecological integrity; intergenerational equity Soil degradation & salinity Productive soil lost faster than it forms Resource-use efficiency; intergenerational equity Fertiliser & nutrient runoff Algal blooms, dead zones in waterways Ecological integrity; user-pays (unpriced pollution) Irrigation Rivers and aquifers drawn down Intragenerational equity (water shared with others)

Food pressure	What happens	Principle challenged
Land clearing for crops/ grazing	Habitat and species lost; carbon released	Ecological integrity; intergenerational equity
Soil degradation & salinity	Productive soil lost faster than it forms	Resource-use efficiency; intergenerational equity
Fertiliser & nutrient runoff	Algal blooms, dead zones in waterways	Ecological integrity; user-pays (unpriced pollution)
Irrigation	Rivers and aquifers drawn down	Intragenerational equity (water shared with others)

05 Pressure 3 — Water

05 Pressure 3 — Water Tiny supply, huge competing demands. About 97.5% of Earth's water is salt water, and most fresh water is locked in ice or deep underground — leaving well under 1% readily accessible for people and ecosystems. That small pool is fought over by three users: agriculture, towns/industry, and the environment itself. When humans take more, the share left for rivers and wetlands (the environmental flow) shrinks. Over-extraction Drawing from rivers and aquifers faster than they refill lowers water tables and dries wetlands. Pollution Nut

06 Pressure 4 — Energy

06 Pressure 4 — Energy Powering development without wrecking the climate. Energy demand rises with both population and rising living standards. The challenge is that the global supply is still dominated by fossil fuels (~80%), whose combustion drives the enhanced greenhouse effect and climate change (you'll meet this in Unit 4). Meanwhile, hundreds of millions of people still lack reliable energy at all — energy poverty — so the world must grow supply and cut emissions simultaneously. The emissions bind Cheap, reliable fossil energy fuelled development

07 Carrying capacity & ecological overshoot

07 Carrying capacity & ecological overshoot When demand grows past what the planet can renew. This second interactive shows the heart of the sustainability problem. The curve is population/demand over time. The dashed line is carrying capacity — what the environment can supply indefinitely. Toggle the scenario: a sustainable path levels off below the line (an "S"/logistic curve); an overshoot path blasts past it, draws down natural capital, and risks collapse. The "Earth's required" gauge translates overshoot into a number students remember. Interactive

08 How the pressures challenge each principle

08 How the pressures challenge each principle The mapping examiners want you to make explicit. A common exam task is to link a pressure to a named sustainability principle and explain the breach. Don't just say "it's bad for the environment" — name the principle and say why it's challenged. EI Conservation of biodiversity & ecological integrity. Clearing land for food, damming rivers for water and mining for energy all remove habitat and disrupt ecosystems. RE Resource-use efficiency. Rising demand makes wasteful use costlier; eroding soil and leaking ir

09Case study — the Boorla Wetlands & Yarrin River

09 Case study — the Boorla Wetlands & Yarrin River All four pressures, in one Victorian catchment. Our running case study makes the abstract concrete. The town of Yarrin sits on the Yarrin River , which feeds the Boorla Wetlands — a Ramsar-listed refuge for migratory birds and a culturally significant place for the local Traditional Owners. Watch how the four pressures collide here: Pressure In the Boorla catchment Effect on the wetland Population Yarrin's population grows; new housing estates approved More household water drawn from the river; more sewage load Food Upstream irrigated dairy & horticulture expand Less river water reaches the wetland; nutrient runoff causes algal blooms Water Drought years cut inflows; extraction licences contested Environmental flow squeezed; wetland edges dry out, bird numbers fall Energy A proposed gas peaker plant vs a community solar farm Cooling water demand (gas) vs land-use change (solar); emissions debate

Pressure	In the Boorla catchment	Effect on the wetland
Population	Yarrin's population grows; new housing estates approved	More household water drawn from the river; more sewage load
Food	Upstream irrigated dairy & horticulture expand	Less river water reaches the wetland; nutrient runoff causes algal blooms
Water	Drought years cut inflows; extraction licences contested	Environmental flow squeezed; wetland edges dry out, bird numbers fall
Energy	A proposed gas peaker plant vs a community solar farm	Cooling water demand (gas) vs land-use change (solar); emissions debate

10Command words decoder

◆ COMMAND WORDS

10 Command words decoder Tap each card. The verb tells you how much thinking the marks expect. Identify / State 1 mk Name it. No explanation needed — e.g. "Water." Don't waste time elaborating. Describe 2–3 mk Give the features or what happens, in order. "Outline what the pressure is and its effect." Explain 2–4 mk Give the why/how — cause and effect. Link the pressure to a consequence with "because...". Analyse 3–6 mk Break it into parts and show relationships — especially the links between pressures and principles. Compare 2–4 mk Give similarities and di

11Four exam traps

⚠ EXAM TRAPS

11 Four exam traps Where good students quietly lose marks. Trap 1 — Treating the four as separate Listing population, food, water and energy as four unrelated bullet points misses the whole point. Show the links. A strategy easing one usually loads another. Trap 2 — Vague "bad for environment" Always name the principle breached and justify it. Generic harm statements don't score. Trap 3 — Confusing numbers with consumption Population pressure is numbers × per-person consumption. A small rich population can out-consume a big poor one. Mention both. Trap 4

12P·E·L writing trainer

P·E·L WRITING

12 P·E·L writing trainer The structure that turns knowledge into marks: Point · Evidence · Link . Most extended-response marks are lost not from missing knowledge but from missing structure . Practise the P·E·L move on this prompt, then reveal a model sentence for each part. Prompt: Explain one way population growth challenges a sustainability principle. 3 marks P Name the pressure and the named principle in one clear sentence. Model: "Population growth challenges intergenerational equity by increasing total demand on finite resources." E Use a concrete



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

13Practice questions

★ PRACTICE & SELF-MARK

13 Practice questions Five VCE-style questions · 22 marks. Compare the weak and strong responses, read the marking guide, then self-mark. Question 1 2 marks Identify two of the four major challenges to upholding the sustainability principles, and for one of them, state a resource it places under pressure. ✗ Weak ✓ Strong Marking guide Weak response The challenges are the environment and pollution. They are bad for nature. Why it loses marks: "environment" and "pollution" aren't the four named challenges (population, food, water, energy). No resource is i

14Key takeaways

♦ RECAP / CHEAT SHEET

14 Key takeaways Four pressures: population, food, water and energy are the named challenges to the sustainability principles. Population is the multiplier — but it's numbers × per-person consumption, not numbers alone. The nexus: food, water and energy are coupled — fixing one often loads another. Always show the links. Carrying capacity & overshoot: when our footprint exceeds biocapacity we draw down natural capital — the definition of unsustainable. Name the principle, then justify. "Harms the environment" scores nothing; "breaches intergenerational e

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The challenges are the environment and pollution. They are bad for nature.

✓ STRONG / MODEL ANSWER

Two challenges are population growth and food production. Food production places fresh water under pressure, as irrigation withdraws large volumes from rivers and aquifers.

✗ WEAK ANSWER

Food is bad because farming destroys things and uses water. This hurts sustainability and the future.

✓ STRONG / MODEL ANSWER

Firstly, clearing native vegetation for cropland removes habitat and reduces species diversity, challenging the conservation of biodiversity and ecological integrity principle. Secondly, intensive irrigation and fertiliser use erode and salinise soil faster than it can reform, depleting a resource future farmers will need — challenging intergenerational equity .

✗ WEAK ANSWER

Food, water and energy are all important and all under pressure. We need to fix all of them to be sustainable.

✓ STRONG / MODEL ANSWER

The three are interdependent, so a fix for one shifts load onto another. Producing more food requires irrigation, increasing water demand, and farm machinery and fertiliser, increasing energy demand. Boosting water supply through desalination is highly energy -intensive, while generating energy from thermal plants needs large volumes of cooling water . Because each lever pulls the others, the challenges form a coupled system — the nexus — and can only be managed together, weighing trade-offs rather than optimised in isolation.

✗ WEAK ANSWER

There are too many people and they use too much, so eventually we run out of resources and it's unsustainable.

✓ STRONG / MODEL ANSWER

An environment has a carrying capacity — the population it can support indefinitely. A population's ecological footprint is the productive land and water it needs to supply resources and absorb wastes. When the global footprint exceeds Earth's biocapacity , humanity is in overshoot : natural capital such as forests, fisheries and aquifers is drawn down faster than it regenerates. Because this depletes the very stocks future supply depends on, growth past carrying capacity cannot continue indefinitely — the definition of unsustainable.

✗ WEAK ANSWER

The town, farms and power station all use the river so the wetland gets less water. This is bad for the birds and the environment. It's hard to be sustainable because everyone wants water. Overall it's a difficult situation.

✓ STRONG / MODEL ANSWER

The three pressures converge on one shrinking resource — the river's environmental flow — making several principles hard to uphold simultaneously. Population: Yarrin's growth raises household water demand and sewage load, challenging ecological integrity (wetland drying) and intragenerational equity (town vs environment). Food: expanding irrigation withdraws more upstream water and adds nutrient runoff, breaching ecological integrity (algal blooms) and the user-pays principle (unpriced pollution). Energy: a gas plant needs cooling water — competing again for flow — and its emissions challenge intergenerational equity and the precautionary principle. Crucially these are linked: any single f